

AVNEESH KUMAR

DevOps & Platform Engineer | GPU Kubernetes | AWS Terraform | Blockchain | 6+ Years

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PROFESSIONAL SUMMARY

Results-driven DevOps & Platform Engineer with 6+ years of experience architecting mission-critical GPU-enabled Kubernetes clusters for Naval defense applications, building offline CI/CD pipelines, and delivering enterprise Hyperledger Fabric blockchain solutions. Proven expertise operating air-gapped RHEL 9-based Kubernetes environments on NVIDIA DGX machines (A100 GPUs), deploying offline Multi-Agent GenAI/LLM systems, and designing cloud-native infrastructure on AWS with Terraform. Strong foundation in full-stack observability (Prometheus/Grafana/Datadog/DCGM), microservices orchestration, and GitOps (ArgoCD). Pursuing CKA certification. Seeking Senior/Staff DevOps or Platform Engineering roles at top-tier organizations.

CORE TECHNICAL SKILLS

Kubernetes & Container Orchestration Kubernetes (GPU-enabled, DGX, RHEL 9), Docker, Containerd, Helm, HPA/VPA Autoscaling, ArgoCD (GitOps), BCM Cluster Management, Offline K8s	CI/CD & DevOps Automation GitLab CI/CD (Offline), Jenkins, GitHub Actions, GitLab Runner, GitOps Principles, Blue-Green & Canary Deployments, Rollback Strategies, Offline Pipelines
Cloud & Infrastructure as Code AWS (EC2, S3, IAM, EKS), Terraform (Modules, Workspaces, Remote State), Multi-region Architecture, Infrastructure Automation, CKA (In Progress)	Observability & Monitoring Prometheus (Custom Metrics), Grafana, Datadog, NVIDIA DCGM & BCM, Alerting & SLOs, GPU Health Monitoring, DGX Cluster Metrics
Blockchain & Distributed Ledger Hyperledger Fabric (Defense/Naval), Ethereum, Solidity, ERC-20/ERC-721/ERC-1155, Hardhat, Truffle, Web3.js, Ether.js, DeFi (Aave, Uniswap)	AI/ML & GenAI (Offline) LLaMA + OpenWebUI (Offline Deployment), Multi-Agent AI Systems, LangChain (RAG), PyTorch, TensorFlow, Hadoop & Spark RAPIDS, GPU Data Pipelines, DALL-E, GPT-3.5
Languages & Databases Python, Solidity, Bash/Shell, JavaScript (ES6), React, PostgreSQL, YAML, HTML/CSS	Storage & Systems DDN Shared Storage, RHEL 9, Linux Administration, Microservices Architecture, SRE Practices, Capacity Planning, Offline Air-Gapped Environments

CERTIFICATIONS & ACHIEVEMENTS

- CKA (Certified Kubernetes Administrator) — In Progress (Target: 30 days)
- GATE 2016 & 2018 Qualified — Top 10% nationally, Computer Science & Engineering
- Hewlett-Packard Education Services — Advanced Technical Training Program
- Nav Bharat Institute of Technology — Certified in C Programming & Software Development
- Team Captain, Cricket & Badminton Championship (B.Tech) — Leadership & Team Management

PROFESSIONAL EXPERIENCE

DevOps Engineer & Blockchain Developer | MicroGenesis Tech Soft Pvt. Ltd. | Dec 2023 – Present | Bengaluru

 **Defense / Naval Domain — Classified Infrastructure**

- Architected and managed 30+ node air-gapped Kubernetes (RHEL 9 / BCM) cluster on NVIDIA DGX A100 machines for mission-critical Naval defense applications, achieving 99.9% uptime across classified networks.
- Built and deployed Hyperledger Fabric private blockchain for secure Naval data synchronization and propagation across 10+ peer organizations in classified offline environments — ensuring tamper-proof data integrity.
- Orchestrated 20+ Naval microservices on the K8s BCM cluster with Dockerization, HPA/VPA autoscaling, and service mesh patterns; managed all cluster lifecycle events including rolling updates and zero-downtime deployments.
- Designed and operated offline CI/CD pipelines using GitLab CE (self-hosted) enabling fully air-gapped build, test, and deploy workflows — reducing deployment cycle time by ~40%.
- Deployed and configured full-stack observability: Prometheus with custom GPU/service metrics, Grafana dashboards, NVIDIA DCGM & BCM for GPU health and utilization tracking across DGX machine clusters — cut P1 incident response time by 60%.
- Built offline Multi-Agent AI platform using OpenWebUI + LLaMA (Dockerized & orchestrated on BCM K8s cluster) with multiple AI agents aligned to Naval project data models and event requirements, reducing documentation lookup time by ~50%.
- Orchestrated Hadoop & Spark RAPIDS workloads on DDN shared storage for GPU-accelerated data processing pipelines serving defense analytics platforms.
- Currently building Terraform AWS modules and ArgoCD GitOps pipelines to extend cloud-native practices; targeting CKA certification within 30 days.

Software Developer – Blockchain | Leeway Hertz Technology | Oct 2022 – Apr 2023 | Remote

- Developed and deployed production Ethereum smart contracts (ERC-20, ERC-721, ERC-1155) for OpenSea-compatible NFT platforms using Hardhat, Truffle, and Ganache.
- Built frontend blockchain interaction layers using Ether.js and Web3.js with seamless MetaMask wallet integration for dApp connectivity.
- Implemented Ethereum Layer-2 scalability solutions (sidechains), reducing end-user gas fees by 70%+.
- Integrated DeFi protocols (Aave, Uniswap) for liquidity management, yield farming, token staking, and on-chain market analysis.
- Managed ICO, IPO, and STO fundraising by designing compliant smart contract-based token issuance mechanisms.

Junior Research Engineer – Blockchain | NIT Goa | Sep 2021 – Sep 2022 | Goa

- Developed end-to-end blockchain infrastructure in Python on Ethereum core protocols, consensus algorithms (PoW, PoS, PoA), and Merkle Tree cryptographic verification.
- Implemented and secured Ethereum smart contracts using advanced cryptography; ensured data immutability across distributed nodes.
- Contributed to research on blockchain protocol optimization; published findings on distributed ledger performance benchmarking.

Python Developer | LT Communication Pvt. Ltd. | Aug 2016 – Dec 2018 | Meerut

- Developed and maintained backend Python applications with MySQL integration, processing high-volume API requests and data pipelines.
- Built RESTful API integrations and backend services, improving system reliability and response performance.
- Contributed to frontend development using HTML, CSS, JavaScript (ES6), and React for responsive client-facing applications.

KEY PROJECTS

Naval Defense Blockchain & AI Platform (Production — Classified)

- Hyperledger Fabric Blockchain: Designed and deployed private blockchain for secure Naval data synchronization and propagation across classified peer networks — ensuring immutability and auditability of mission-critical data.

- GPU K8s Microservices Platform: Orchestrated all Naval project microservices on RHEL 9 BCM Kubernetes cluster on NVIDIA DGX A100; configured HPA autoscaling, Prometheus/Grafana monitoring, and GitLab CE CI/CD pipelines in fully offline environment.
- Offline Multi-Agent AI System: Built and deployed offline AI platform (OpenWebUI + LLaMA) on BCM K8s cluster with multiple AI agents trained on Naval data models and project-specific event requirements.

Capstone: Full-Stack Cloud-Native Infrastructure (GitHub Portfolio — In Progress)

- Building complete IaC demo: EKS cluster on AWS via Terraform + ArgoCD GitOps + Prometheus/Grafana observability stack — public GitHub repo for interview demonstration at top-tier companies.

INSTINCT Games – NFT Marketplace (Ethereum)

- Developed Ethereum smart contracts (ERC-721 & ERC-1155) with buy/sell, deposit/withdraw, and royalty distribution; deployed on testnets with Ether.js and MetaMask integration.

Polpappa – Decentralized Crypto Exchange (DeFi)

- Architected hack-resistant DEX integrating Uniswap and Aave for automated market making, lending/borrowing, and NFT marketplace modules within a Metaverse ecosystem.

Redgore InteriorAI / LGPT – Generative AI & LLM

- Fine-tuned generative AI models (GPT-3.5, DALL-E, Stable Diffusion, LLaMA) within PyTorch/TensorFlow; integrated LangChain for RAG pipelines powering intelligent design recommendation systems.

Smart Microgrid Energy Management – M.Tech Research

- Designed blockchain-powered smart controller using DLT for energy utilization optimization; implemented smart contracts for demand prediction, data immutability, and metering visualization.

EDUCATION

M.Tech – Computer Science & Engineering | NIT Durgapur | 2019–2021 | CGPA: 7.08/10

Thesis: Skin/Scalp Disease Recognition & Human Activity Recognition using Deep Learning; Triage System using ML

B.Tech – Computer Science & Engineering | SCRIET, CCS University, Meerut | 2012–2016 | CGPA: 7.5/10

TOOLS & DEVELOPMENT ENVIRONMENTS

IDEs: VS Code, Remix IDE **Blockchain:** Hardhat, Truffle, Ganache, Hyperledger Fabric **DevOps:** GitLab CE, Jenkins, Helm, ArgoCD, Prometheus, Grafana, Datadog, NVIDIA DCGM/BCM **Cloud:** AWS (EC2, S3, IAM, EKS), Terraform **AI/ML:** OpenWebUI, LLaMA, LangChain, PyTorch, TensorFlow, Spark RAPIDS

LANGUAGES

English (Professional Proficiency) | Hindi (Native)